Testing Web Applications *Using* Open Source Tools

Before a software application or a Web application can be released to the public, it has to undergo rigorous testing procedures to ensure its reliability, scalability, durability and its performance under load or stress. This article gives readers an insight into the types of testing required to evaluate the robustness of the software or Web application, and gives an overview of open source tools that can be used for the purpose.

The process of software development integrates the tasks of rigorous testing on multiple parameters and dimensions so that the overall performance of the software suite is good. Software testing is mandatory for every type of software product, whether it is for the desktop, is standalone or a Web based application. Without proper testing, the software application can behave abnormally or can crash when subjected to specific inputs.

Testing strategies for Web applications

***Load testing*:** This is the process of evaluating the behaviour of software being accessed by a large number of users concurrently. The capacity of the software to handle the load of users is analysed.

Stress testing: This analyses the robustness of software. It identifies the specific points at which the software modules incur problems, and evaluates them under extreme conditions of system failure.

Security testing: This discovers the vulnerabilities and security loopholes in the software. It also includes penetration testing, whereby one tries to identify the hacking or cracking probabilities of the software.

Smoke testing: This ensures that important and key features of the software product are working well. It is used to check the stability of software for further stages of testing. The most critical functions are analysed in this testing strategy.

Unit testing: The software is tested at the component and module levels so that the behaviour of individual units can be analysed.

Acceptance testing: The evaluation of software based on the prescribed software requirements specifications (SRS) is done to make it delivery-ready. The levels and scores for the acceptability of the software product are investigated.

***Graphical user interface (GUI) testing*:** The proper functioning of the GUI or simply the front-end of the application is tested to ensure the software is user friendly and able to function as per the requirements.

***Gorilla testing*:** The software modules are tested repeatedly under assorted scenarios and inputs so that their consistency can be checked. This is also referred to as frustrating testing because it involves iterative testing of the same component and the identification of bugs.

***Performance testing*:** This involves evaluating the software product on multiple aspects including speed, load, traffic, stress, susceptibilities and many others.

Prominent tools for Web application testing

Table 1 lists open source tools for Web app testing.

Locust

URL: *https://locust.io/*

Locust is a Python based tool to evaluate the behaviour of a Web application that is being used by multiple concurrent users. Locust can generate the traffic that connects to the Web application in order to test its behaviour under huge load. It is one of the high performance, free and open source tools for load testing. Locust defines the behaviour and load of virtual traffic on the Web application or server. It creates enormous, swarm based traffic so that a random load can be put on the website.

For example, to test the behaviour of a website with 1000 concurrent users, you might actually need to get those 1000 users, which in reality is practically impossible. Using Locust, any number of users can be generated and a report based